KSJ RESEARCH

Research Note Vol.1

Is AI Power Demand Real, and Does the Market Already Know?

Demand, ground truth, earnings and market pricing from official data — and what it means for IPO pricing

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Demand is real, it shows up on the ground — the market already knows, and multiples ran ahead of earnings (author's view)

IEA 2030E 945 TWh (2.3x 2024) · Virginia commercial sales +73% vs U.S. +10% · 4 names' market cap 7.7x combined (range 4.4x–26.1x, simple average 14.6x)

Key metrics [*1,*2,*3,*4,*5]

Key metric	Value	Page
2030E data-centre power	**945 TWh (2.3x)**	1
2035E outlook range	700~1,700 TWh	1
2025 growth: AI DC · DC · world	+50% · +17% · +3%	1
Virginia commercial index (2025)	**173 vs U.S. 110**	2
4 names' combined cap (2023-01→2026-10)	**7.7x**	4
Simple average / range	14.6x / 4.4x~26.1x	4
Cap x > NI x (FY2023A→FY2025A)	**4 of 4**	3·4
Earnings · multiple (FY2022A base)	after FY2022	4
Current P/E (FY2025A)	agg. 64.2x · median 60.4x	4
KTB 10y (start-2023→latest)	3.81 → 4.21%	5

Source: *1 IEA, Energy and AI (Apr 2025), executive summary *2 IEA, Key Questions on Energy and AI (Apr 2026) *3 U.S. Energy Information Administration (EIA), Electricity retail sales *4 Korea Exchange (KRX) Open API, KOSPI daily trading *5 FSS DART annual reports (consolidated)
Note: KSJ Research calculations; see footnotes on each page

2.3x

Data-centre power 2030E ÷ 2024 (IEA Base Case)

Demand is real — 945 TWh by 2030; AI data-centre demand +50%

- 2025: data centres +17%, AI data centres +50% vs world +3%
- Grew 12%/yr in 2017–2024; 1.5% of world power in 2024; U.S. 45%
- But the 2035 range is wide: 700–1,700 TWh (±42% vs base)

+73%

Virginia commercial power sales 2015→2025 (EIA)

It shows up on the ground — 63 index points ahead of the U.S. (+10%)

- CAGR Virginia 5.6% vs U.S. 0.9% (2015–2025)
- Last 3 years (2022→2025) CAGR 6.8% vs 2.4%
- U.S. commercial price +26% since 2015 (volume +9.8%)

7.7x

4 names, combined cap, 2023-01-02→2026-10-19 (range 4.4x–26.1x)

The market already knows — for 4 of 4, market cap outgrew earnings (author's view)

- Simple average 14.6x — Doosan Enerbility (4.4x), 72% of starting cap, drags the total
- Current P/E aggregate 64.2x, median 60.4x (latest cap ÷ FY2025A NI)
- NI of 4 names FY2023A→FY2025A 2.6x vs cap 7.7x; FY2022-based split after collection

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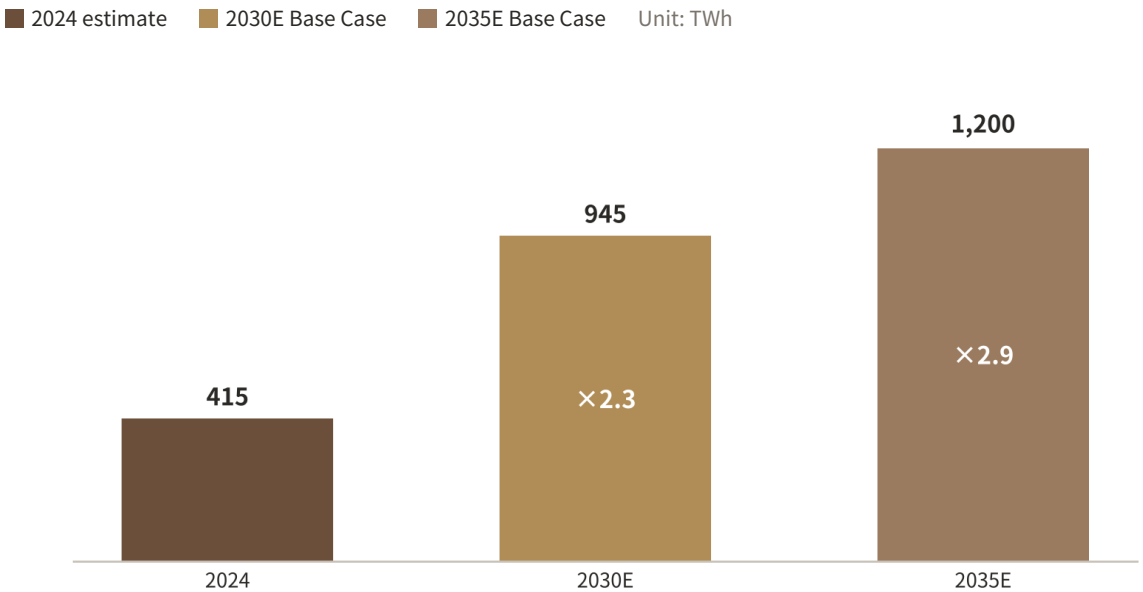
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Demand is real: data-centre power use reaches 2.3x by 2030; AI data-centre demand +50% in 2025

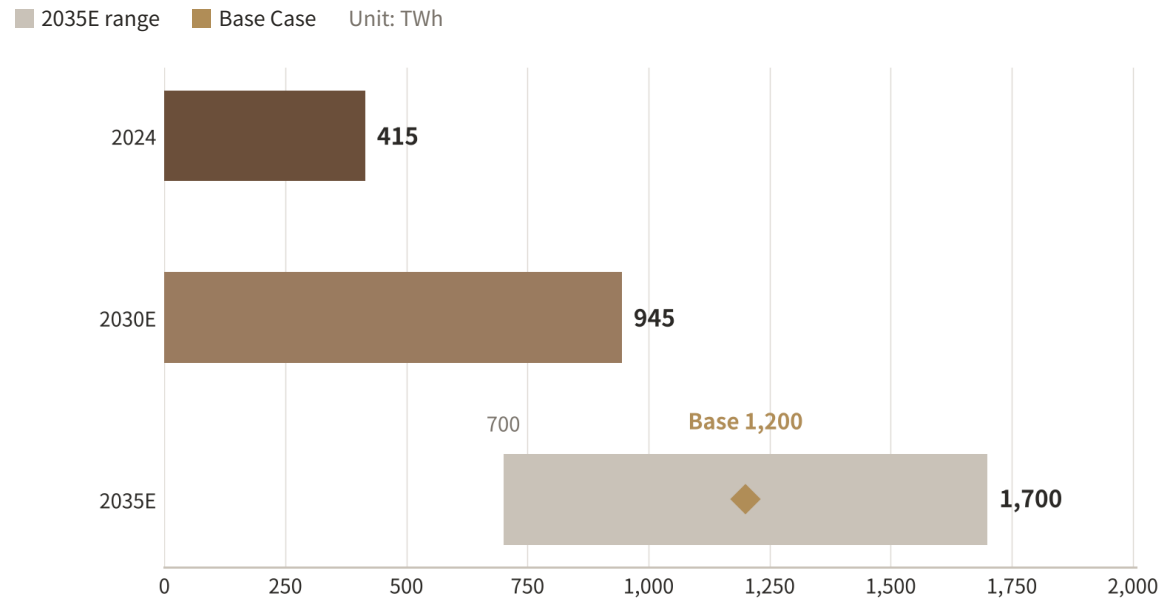
IEA Base Case: 415 TWh (2024) → 945 TWh (2030) → 1,200 TWh (2035) — but the 2035 range (700–1,700 TWh) is wide

Data-centre power use 2030E 945 TWh — 2.3x 2024 *1



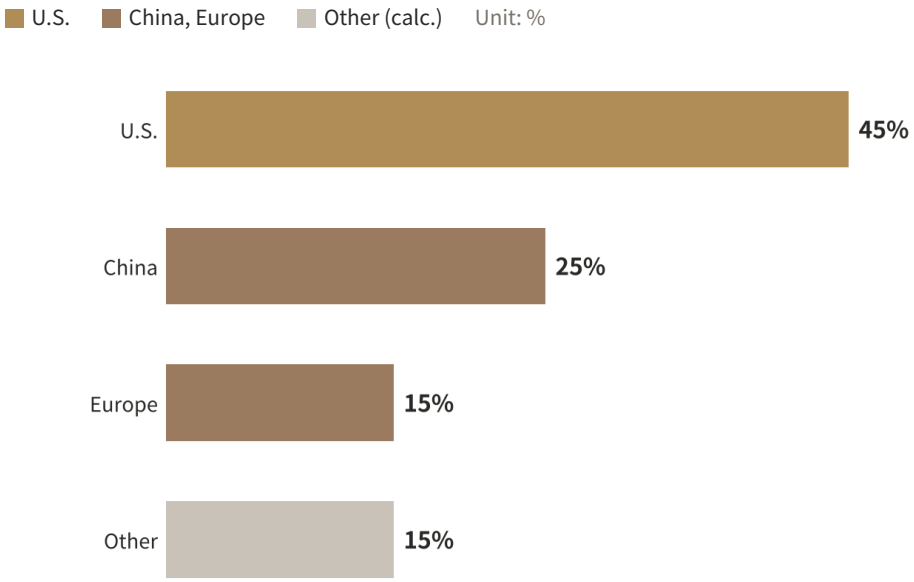
Source: *1 IEA, Energy and AI (Apr 2025), executive summary
Note: KSJ Research. Multiples inside bars = year ÷ 2024

2035 range 700–1,700 TWh — −42% to +42% vs Base Case *1



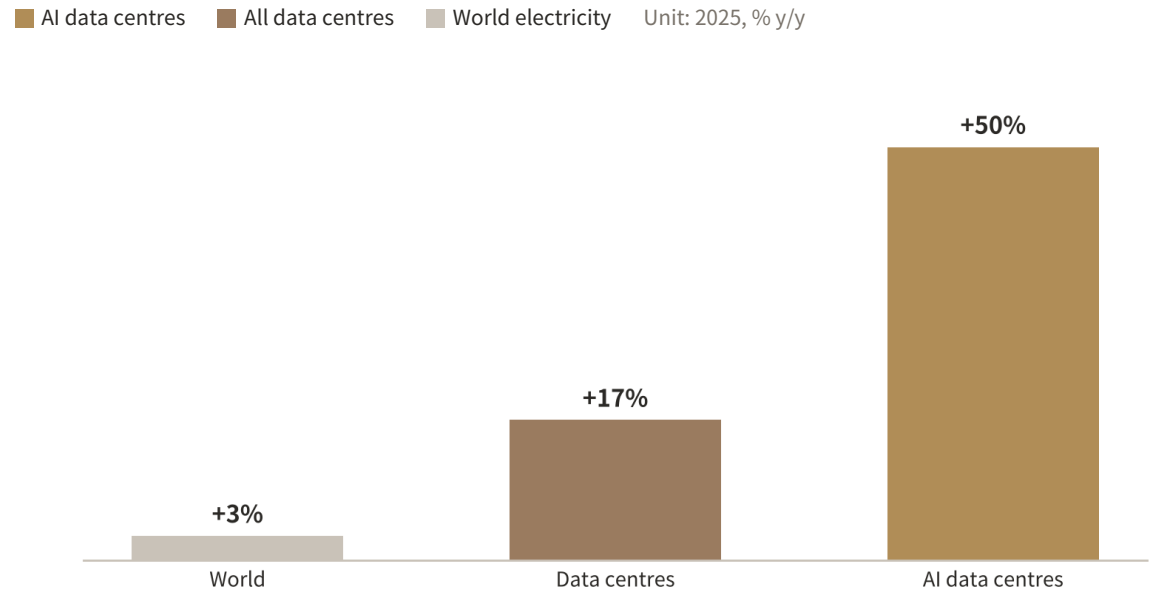
Source: *1 IEA, Energy and AI (Apr 2025), executive summary
Note: Range = IEA's 2035 low–high; % vs Base Case (KSJ Research)

The U.S. used 45% of data-centre power in 2024; U.S., China and Europe 85% *1



Source: *1 IEA, Energy and AI (Apr 2025), executive summary
Note: Other = 100 − U.S., China, Europe (KSJ Research). Data-centre power grew 12%/yr in 2017–2024; 1.5% of world power in 2024

AI data-centre demand +50% in 2025 — 17x the pace of world power (+3%) *1

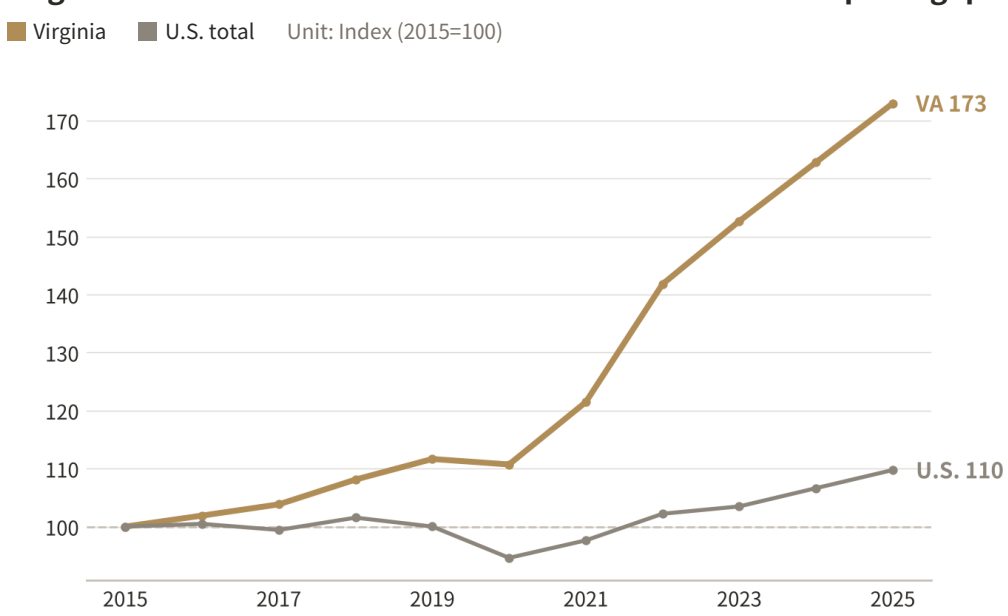


Source: *1 IEA, Key Questions on Energy and AI (Apr 2026)
Note: Multiple = 50 ÷ 3 (KSJ Research)

It shows up on the ground: Virginia commercial sales +73% since 2015, 63 index points ahead of the U.S. (+10%)

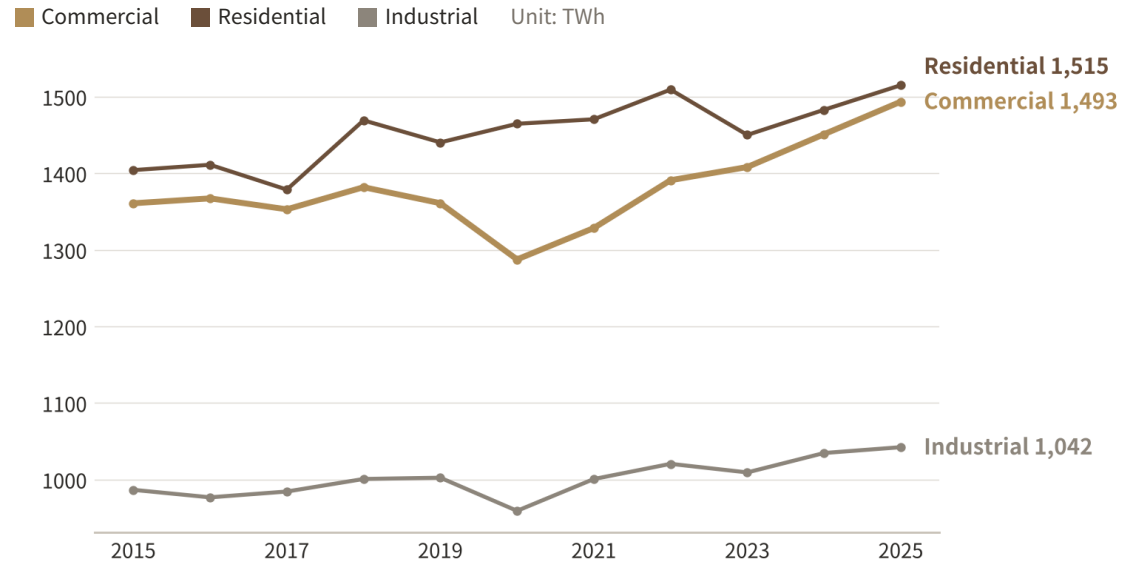
Virginia, known as a data-centre hub, grew commercial sales 4.7pp a year faster than the U.S. — U.S. commercial prices rose +26%

Virginia commercial sales +73% vs U.S. +10% — 63-index-point gap in 2025 *1



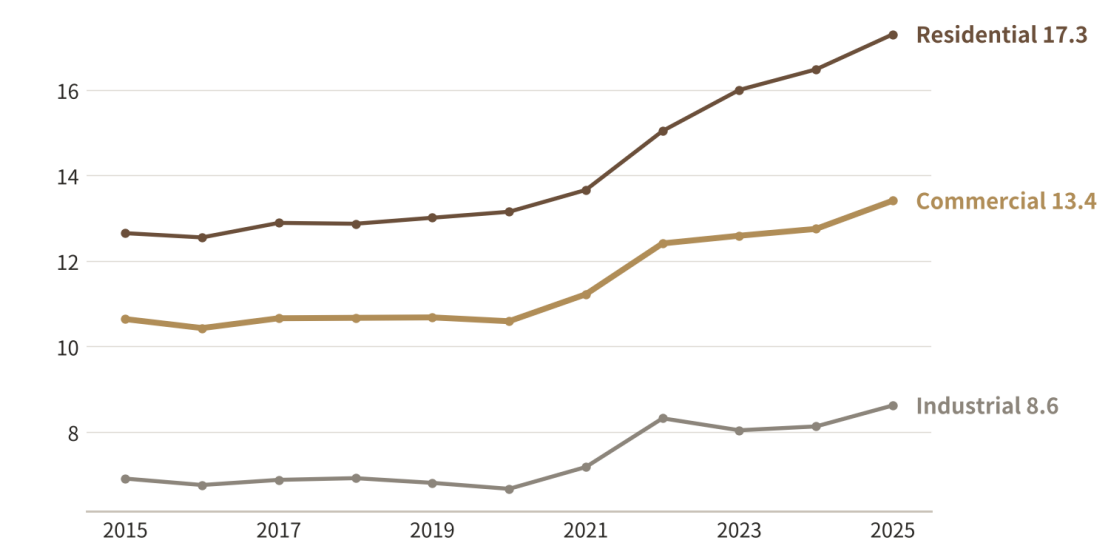
Source: *1 U.S. Energy Information Administration (EIA), Electricity retail sales
Note: Index = sales ÷ 2015 × 100 (KSJ Research). CAGR Virginia 5.6% vs U.S. 0.9%

U.S. sales by sector 2015→2025: Commercial +9.8% · Residential +7.9% · Industrial +5.6% *1



Source: *1 U.S. Energy Information Administration (EIA), Electricity retail sales
Note: Sales in million kWh ÷ 1,000 = TWh (KSJ Research). Includes the 2020 commercial dip

Commercial average price 10.64¢ → 13.41¢ (+26%) — faster rise since 2020 *1



Source: *1 U.S. Energy Information Administration (EIA), Electricity retail sales
Note: Commercial price CAGR 2015–2020 -0.1% → 2020–2025 4.8% (KSJ Research)

Last 3 years CAGR: Virginia 6.8% vs U.S. 2.4% *1

Unit: % y/y; gap in pp

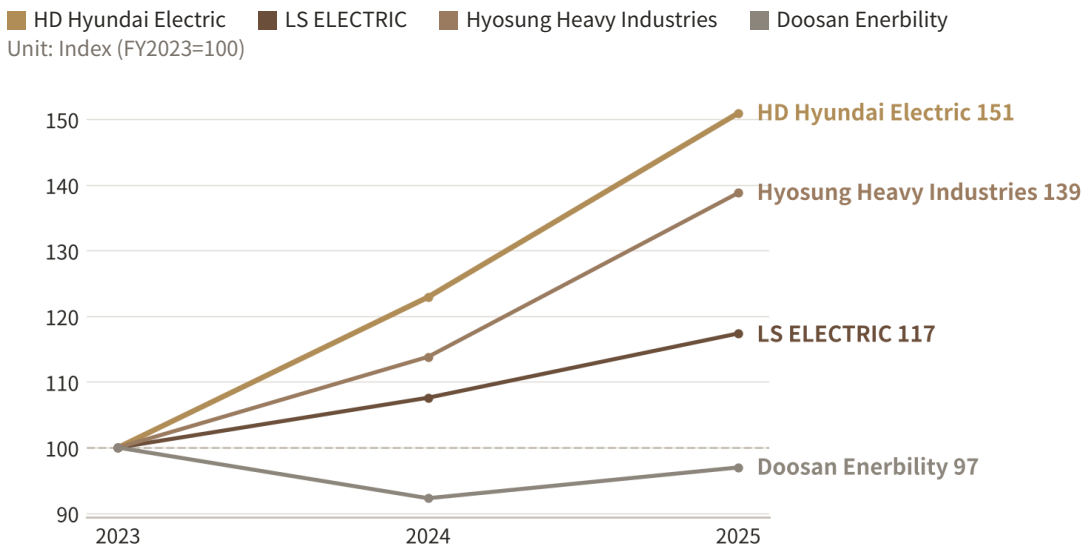
Year	Virginia	U.S. total	Gap
2018	+4.1%	+2.1%	+2.0%p
2019	+3.3%	-1.5%	+4.8%p
2020	-0.8%	-5.4%	+4.6%p
2021	+9.7%	+3.2%	+6.5%p
2022	+16.7%	+4.7%	**+12.0%p**
2023	**+7.7%**	+1.2%	+6.4%p
2024	**+6.6%**	+3.0%	+3.6%p
2025	**+6.2%**	+2.9%	+3.3%p

Source: *1 U.S. Energy Information Administration (EIA), Electricity retail sales
Note: Commercial sales growth y/y (KSJ Research). Shaded = last 3 years; gold = largest gap (2022)

Earnings grew too: NI to parent of 4 names 2.6x from FY2023A to FY2025A — but slower than market cap (7.7x)

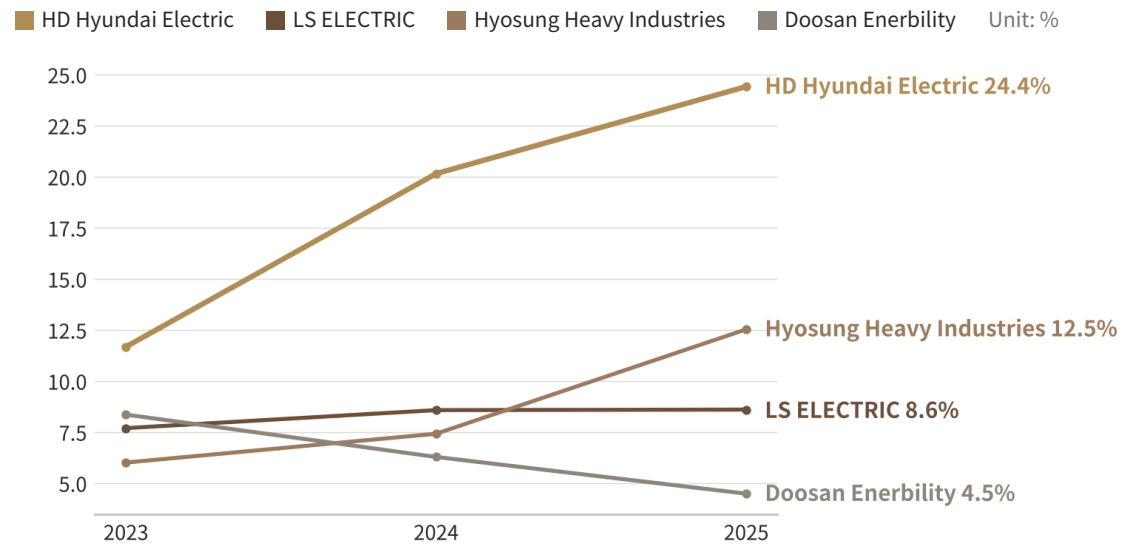
Combined revenue of 4 names +11%, operating profit of 4 names +24% (FY2023A→FY2025A, consolidated)

Revenue index (FY2023=100): HD Hyundai Electric highest at 151 · Doosan Enerbility 97 — combined +11% *1



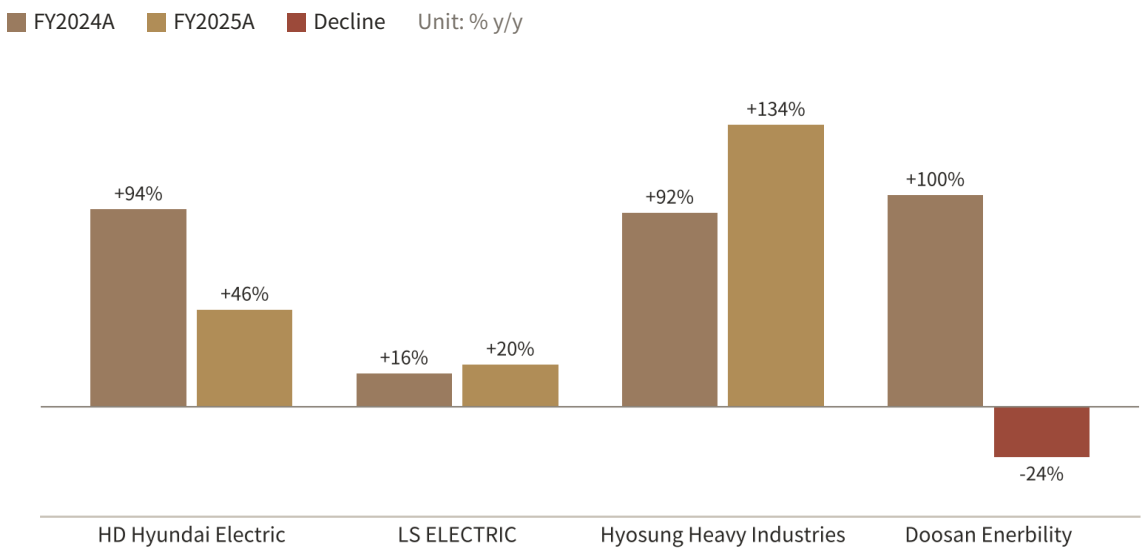
Source: *1 FSS DART annual reports (consolidated)
Note: Index = revenue ÷ FY2023A × 100 (KSJ Research). Consolidated; absolute values in the FY2025A table. FY2022A added after collection (python -m research.vol01.fetch_extra --only dart)

Operating margin FY2025A: HD Hyundai Electric highest at 24.4%; 3 of 4 improved *1



Source: *1 FSS DART annual reports (consolidated)
Note: Operating margin = operating profit ÷ revenue (KSJ Research). FY2022A added after collection (python -m research.vol01.fetch_extra --only dart)

NI to parent FY2023A→FY2025A: 4 names combined 2.6x; Hyosung Heavy Industries leads at 4.5x *1



Source: *1 FSS DART annual reports (consolidated)
Note: NI attributable to owners of the parent; no growth rate after a loss year (no bar). FY2022A added after collection (python -m research.vol01.fetch_extra --only dart)

FY2025A combined: revenue 320,717, OP 29,314, NI 16,239 (KRW 100mn) *1

Unit: KRW 100mn, %, x

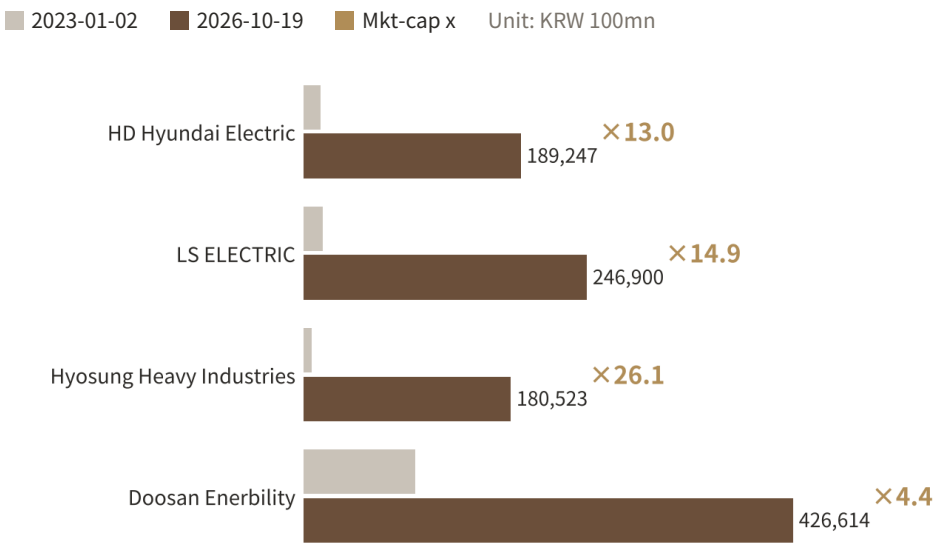
Company	Revenue	OP	OPM	NI FY2023	NI FY2025	NI x
HD Hyundai Electric	40,795	9,953	24.4%	2,592	7,326	2.8x
LS ELECTRIC	49,658	4,264	8.6%	2,060	2,866	1.4x
Hyosung Heavy Industries	59,685	7,470	12.5%	1,160	5,199	**4.5x**
Doosan Enerbility	170,579	7,627	4.5%	556	848	1.5x
Total	**320,717**	**29,314**	**9.1%**	**6,367**	**16,239**	**2.6x**

Source: *1 FSS DART annual reports (consolidated)
Note: Revenue, OP and OPM are FY2025A consolidated. NI x = FY2025A ÷ FY2023A; a label replaces the multiple when the base is ≤0 (KSJ Research)

The market has already moved: 4 names' market cap 7.7x combined (range 4.4x–26.1x, simple average 14.6x)

Even on available earnings (FY2023A→FY2025A), 4 of 4 saw market cap outgrow NI — FY2022-based split after collection

Market cap: 7.7x combined · 14.6x simple average — 4.4x to 26.1x by name [*1,*2]



Source: *1 Korea Exchange (KRX) Open API, KOSPI daily trading *2 Company filing (stock split), author's check
Note: Market cap = close × listed shares at each date (KRX MKTCAP). The combined multiple is below the average because Doosan Enerbility (4.4x) was 72% of the starting cap. LS ELECTRIC: 5:1 split on 2026-04-13 — we use the market-cap multiple 14.9x, not the raw price multiple 3.0x (split-adjusted 14.9x).

Market-cap change split: earnings vs multiple (FY2022A base) [*1,*2]

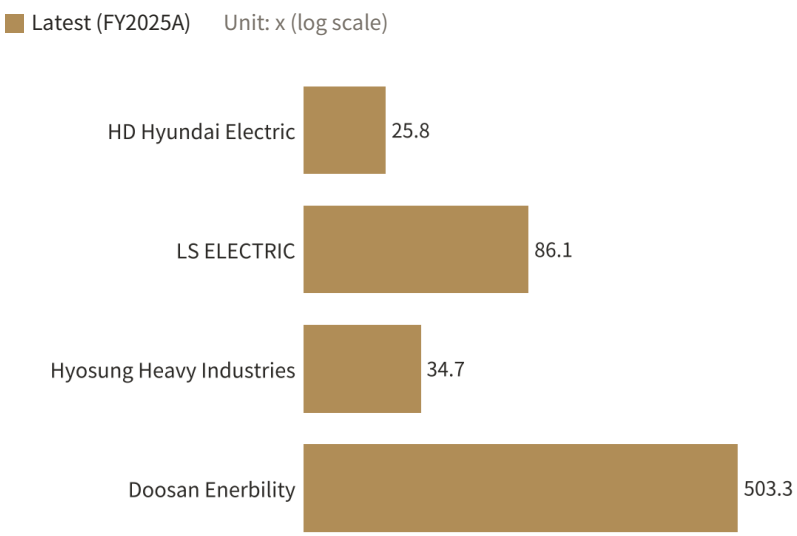
Generated after data collection

Reason: FY2022A NI to parent missing (pairs with start-2023 market cap)

```
python -m research.vol01.fetch_extra --only dart
```

Source: *1 Korea Exchange (KRX) Open API, KOSPI daily trading *2 FSS DART annual reports (consolidated)
Note: Log split Δln(cap) = Δln(NI) + Δln(P/E). Start-2023 cap ↔ FY2022A, latest cap ↔ FY2025A NI to parent. Only names with positive NI in both years are split (KSJ Research)

Current P/E: aggregate 64.2x · median 60.4x — Doosan Enerbility 503x reflects thin earnings [*1,*2]



Source: *1 Korea Exchange (KRX) Open API, KOSPI daily trading *2 FSS DART annual reports (consolidated)
Note: P/E = latest cap ÷ FY2025A NI (KSJ Research). Log scale. Start-2023 P/E (FY2022A) shown after collection (python -m research.vol01.fetch_extra --only dart)

Split by company (FY2022A base) [*1,*2]

Generated after data collection

Reason: FY2022A NI to parent missing (pairs with start-2023 market cap)

```
python -m research.vol01.fetch_extra --only dart
```

Source: *1 Korea Exchange (KRX) Open API, KOSPI daily trading *2 FSS DART annual reports (consolidated)
Note: Columns: cap x · NI x · P/E · earnings share · multiple share

Hence Vol.2: how to price an IPO in a sector whose multiples have re-rated (author's view)

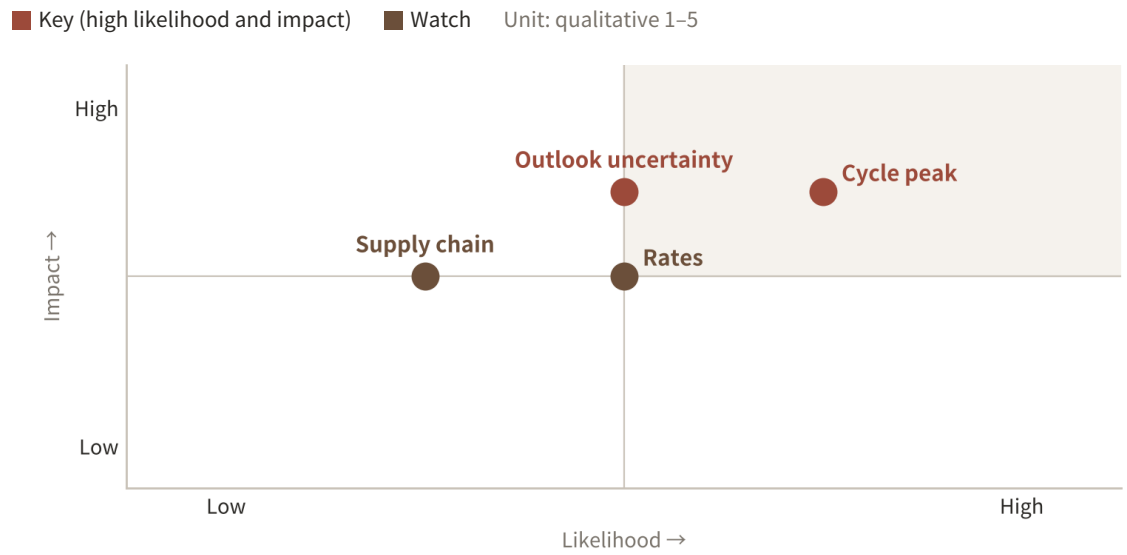
Implications through three lenses (ECM pricing · data-centre PF · capacity options) and four risks

Three lenses: data → judgement *1,*2,*3,*4

Lens	What the data says	Implication (author's view)
ECM pricing	current P/E aggregate 64.2x · median 60.4x (FY2025A); cap 7.7x combined (range 4.4x–26.1x, simple average 14.6x)	When peer multiples have run ahead of earnings, defend the price with conservative earnings and an IPO discount rather than raw peer P/E
Data-centre PF	Virginia commercial sales 5.6%/yr; U.S. commercial price +26% (2015→2025)	Where volume and price rise together, PPAs and grid-connection timing are the key PF cash-flow variables
Capacity options	Revenue +11% (FY2023A→FY2025A); top OPM 24.4%; IEA 2035 range 700–1,700 TWh	With a wide 2035 range, phased capacity additions that expand as demand is confirmed are preferable

Source: *1 Korea Exchange (KRX) Open API, KOSPI daily trading *2 FSS DART annual reports (consolidated) *3 U.S. Energy Information Administration (EIA), Electricity retail sales *4 IEA, Energy and AI (Apr 2025), executive summary
Note: Figures in the data column come from pages 1–4

Risk matrix: cycle peak and outlook uncertainty matter most (author's view) *1,*2,*3



Source: *1 IEA, Energy and AI (Apr 2025), executive summary *2 Bank of Korea ECOS market rates (daily) *3 FRED, Federal Reserve Bank of St. Louis (DGS10)
Note: Anchors: IEA 2035 range ±42% vs base · current P/E aggregate 64.2x · median 60.4x (FY2025A) · KTB 10y 3.81→4.21% · UST 10y 3.79→5.09% (2023-01-02→2026-10-16) · supply chain is qualitative. Positions are the author's judgement (config ASSUMPTION)

Next in Vol.2: price range for a hypothetical unlisted power-equipment IPO

How to price an IPO in a sector whose multiples have re-rated

- Set up Target Co. — every business and financial assumption disclosed
- Peer P/E and EV/EBITDA: 4 Korean names + overseas peers (after SEC resumes)
- Fair value → IPO discount → indicative price range
- Monte Carlo sensitivity of the range

Note: Timing and method of Vol.2 may change

Limitations of this note

- Ground data covers commercial sales only; data-centre load may be classified as industrial
- IEA figures are summary values entered manually and checked against the source
- The market window (Jan 2023–Oct 2026) and earnings window (FY2022A–FY2025A) do not match exactly
- P/E uses reported NI without one-off adjustments; thin earnings distort P/E
- The sample is four Korean names; overseas peers (SEC) are in progress

Note: To be addressed from Vol.2 onward

Expected Q&A: answered with numbers

Questions likely in interviews or due diligence, with page references

8 expected questions [*1,*2,*3,*4,*5]

Question	Answer	Page
Isn't AI power demand overhyped?	IEA: 945 TWh in 2030E (2.3x 2024); AI data centres +50% in 2025 vs world +3%. The 2035 range (700–1,700 TWh) is wide	1
Isn't Virginia a special case?	It is one state. But commercial sales are +73% since 2015 (U.S. +10%) and the gap is widening (6.8%/yr over 3 years). Industrial classification is a stated limitation	2, 5
Do earnings support it?	NI of 4 names 2.6x FY2023A→FY2025A — growing, but slower than market cap (7.7x)	3, 4
Has the market priced it in?	Cap 7.7x combined (range 4.4x–26.1x, simple average 14.6x); on available earnings, 4 of 4 saw cap outgrow NI (author's view). FY2022-based split after collection	4
Why is LS ELECTRIC's price up 3x but its market cap 15x?	A 5:1 split on 2026-04-13. The price move KRW 55,100 → 164,600 (2.99x) compares different share sizes; split-adjusted 2.99×5 = 14.9x, matching the cap multiple (shares 30,000,000→150,000,000)	4, A2
Which is right: 7.7x combined or 14.6x average?	Both — they answer different questions. Combined is a cap-weighted basket; the average weights names equally. Doosan Enerbility (4.4x) was 72% of the starting cap, pulling the combined figure down. Use combined for sector value, average for per-name re-rating	4
P/Es are high — is it a bubble?	current P/E aggregate 64.2x · median 60.4x (FY2025A). Doosan Enerbility's 503x reflects thin NI (848, KRW 100mn). It depends on earnings durability (author's view)	4, 5
How can this be reproduced?	python -m src.data_fetch.fetch_all → python -m research.vol01.fetch_extra → python -m research.vol01.build. Code computes every figure from data/ files	A3

Source: *1 IEA, Energy and AI (Apr 2025), executive summary *2 U.S. Energy Information Administration (EIA), Electricity retail sales *3 Korea Exchange (KRX) Open API, KOSPI daily trading *4 FSS DART annual reports (consolidated) *5 Company filing (stock split), author's check
Note: Numbers match the body; shaded = data-treatment questions

Data notes: sources, APIs, dates and the KRX market-cap check

SEC in progress · LS ELECTRIC split treatment stated · market cap uses listed shares at each date

Data sources · APIs · as-of · status

Source	API	Items	As of	Status
DART · FSS	Open DART fnlttSinglAcntAll (annual report 11011, consolidated first)	Revenue, OP, NI to parent (4 peers)	2026-10-20	Collected
KRX · Korea Exchange	KRX Open API stk_bydd_trd (KOSPI daily trading)	Close, market cap, listed shares	2026-10-20	Collected
EIA · U.S. EIA	EIA API v2 electricity/retail-sales	Retail sales and price by state/sector	2026-10-20	Collected
ECOS · Bank of Korea	ECOS StatisticSearch 817Y002 (daily market rates)	KTB 3y/10y, USD/KRW	2026-10-20	Collected
FRED · St. Louis Fed	FRED series/observations	DGS10, DEXKOUS	2026-10-20	Collected
SEC EDGAR	data.sec.gov companyfacts (XBRL)	Overseas peers' financials	—	In progress
IEA reports	Manual entry (Energy and AI Apr 2025, Key Questions Apr 2026)	Data-centre power outlook and growth	2026-10-21	Verified vs source

Note: As-of = as_of in each .meta.json (IEA: verification date). Raw API data (data/raw) is not committed; processed and manual files are. Repo: https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio

KRX market-cap check: listed shares at each date (2023-01-02 · 2026-10-19) *1,*2

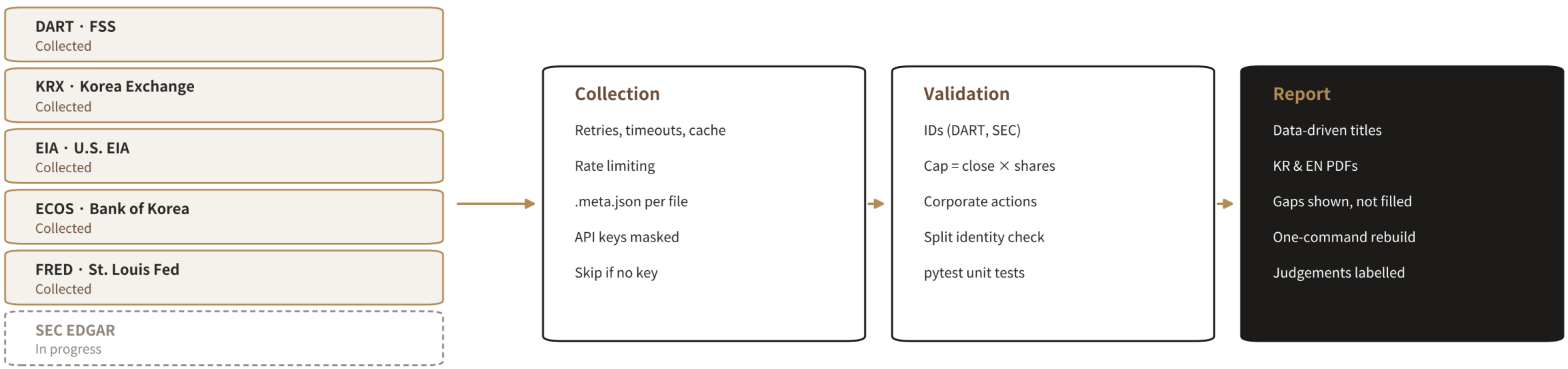
Company	Shares 2023-01-02	Shares 2026-10-19	Corporate action	Price x (raw)	Split-adj. price x	Cap x	Cap check
HD Hyundai Electric	36,047,135	36,047,135	No change	12.98x	12.98x	12.98x	0.00%
LS ELECTRIC	30,000,000	150,000,000	5:1 split 2026-04-13 (ratio matches)	2.99x	14.94x	**14.94x**	0.00%
Hyosung Heavy Industries	9,324,548	9,324,548	No change	26.06x	26.06x	26.06x	0.00%
Doosan Enerbility	638,308,033	640,561,146	Shares +0.35% — no recorded action; check filings	4.37x	4.37x	4.38x	0.00%

Source: *1 Korea Exchange (KRX) Open API, KOSPI daily trading *2 Company filing (stock split), author's check
Note: Cap = KRX MKTCAP (close × listed shares at each date). Check = close × shares ÷ MKTCAP − 1. Split-adjusted price x = price x × split ratio. Price charts must use split-adjusted prices. Actions are recorded in data/manual/krx_corporate_actions.csv (author's check)

What I built: 6 official sources, collected and validated automatically into this report

Code computes every number from data files — missing data stays a grey box

6 official sources → collection → validation → report [*1]



Source: *1 KSJ Research
Note: 5/6 collected (SEC in progress). IEA figures entered manually and verified

Reproduce (local)

https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio

- pip install -r requirements.txt
- python -m src.data_fetch.fetch_all --source all --asof 2026-10-20
- python -m research.vol01.fetch_extra --asof 2026-10-20
- python -m research.vol01.build
- pytest -q
- Keys only from .env; raw data (data/raw) not committed

Note: Run from the repository root

Glossary: units and calculations used in this note

Units: TWh, KRW 100mn, %, x

Glossary

Term	Definition
TWh	Terawatt-hour, one billion kWh; annual electricity use
Base Case	IEA's central scenario; the range is IEA's low–high
Commercial sector	EIA retail-sales class; data centres may be commercial or industrial by state
Index (2015=100)	Year value ÷ 2015 value × 100
CAGR	Compound annual growth = (end ÷ start)^(1/years) − 1
Market cap	Close × listed shares at that date (KRX MKTCAP)
Combined multiple	Sum of caps (end) ÷ sum (start); cap-weighted
Simple-average multiple	Average of each name's multiple; equal-weighted
Stock split	One share becomes several; cap unchanged, price adjusts → compare split-adjusted prices
NI to parent	Net income attributable to owners of the parent (DART consolidated)
PER	Market cap ÷ NI to parent
Log decomposition	Δln(cap) = Δln(NI) + Δln(P/E), an identity since cap = NI × P/E
Earnings / multiple share	Each term ÷ Δln(cap)
ECM	Equity capital markets: IPOs, follow-ons
Indicative price range	Price range set before book-building
PF	Project finance, repaid from project cash flows

Note: Shaded = core calculation terms in this note

Disclosures

- This is a personal portfolio for education and research. It is not investment advice or a recommendation to buy or sell any security.
- Views are the author's own and do not represent any current or future employer.
- All data are public; accuracy and completeness are not guaranteed. Investors are solely responsible for their decisions.
- KSJ Research is a personal research brand and is not affiliated with any financial institution.
- As of 2026-10-20. Figures, sources and methods can be reproduced from Appendix A2 and the code.

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https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio